10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 5.4 — Equations Reducible to Quadratic Form

Unit 5 | Solved Exercise | New Book (PECTAA)

Exercise 5.4 (Solved)

These equations are not quadratic, but a suitable substitution turns them into a quadratic equation. After solving for the new letter, go back to the original letter and solve again. Reject the values that do not fit.

1. Solve the following equations.

(i) $5x^4 - 19x^2 + 12 = 0$

Put $y = x^2$

Then $x^4 = y^2$

The equation becomes $5y^2 - 19y + 12 = 0$

$5y^2 - 15y - 4y + 12 = 0$

$5y(y - 3) - 4(y - 3) = 0$

$(5y - 4)(y - 3) = 0$

$5y - 4 = 0$ or $y - 3 = 0$

$y = 4/5$ or $y = 3$

Put back $y = x^2$

$x^2 = 4/5$ or $x^2 = 3$

$x = \pm 2/\sqrt{5}$ or $x = \pm \sqrt{3}$

$2/\sqrt{5} = 2\sqrt{5}/5$

Answer: $x = \pm \sqrt{3}$, $x = \pm 2\sqrt{5}/5$

(ii) $4x^4 - 27x^2 + 18 = 0$

Put $y = x^2$

Then $x^4 = y^2$

The equation becomes $4y^2 - 27y + 18 = 0$

$4y^2 - 24y - 3y + 18 = 0$

$4y(y - 6) - 3(y - 6) = 0$

$(4y - 3)(y - 6) = 0$

$y = 3/4$ or $y = 6$

Put back $y = x^2$

$x^2 = 3/4$ or $x^2 = 6$

$x = \pm \sqrt{3}/2$ or $x = \pm \sqrt{6}$

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Answer: $x = \pm \sqrt{6}, x = \pm \sqrt{3}/2$

(iii) $5x^4 - 22x^2 + 8 = 0$

Put $y = x^2$

Then $x^4 = y^2$

The equation becomes $5y^2 - 22y + 8 = 0$

$5y^2 - 20y - 2y + 8 = 0$

$5y(y - 4) - 2(y - 4) = 0$

$(5y - 2)(y - 4) = 0$

$y = 2/5$ or $y = 4$

$x^2 = 2/5$ or $x^2 = 4$

$x = \pm \sqrt{10}/5$ or $x = \pm 2$

Answer: $x = \pm 2, x = \pm \sqrt{10}/5$

(iv) $4 \cdot 2^{2x+1} - 9 \cdot 2^x + 1 = 0$

Write $2^{2x+1} = 2 \cdot (2^x)^2$

So $4 \cdot 2^{2x+1} = 8(2^x)^2$

$8(2^x)^2 - 9(2^x) + 1 = 0$

Let $2^x = y$

$8y^2 - 9y + 1 = 0$

$8y^2 - 8y - y + 1 = 0$

$8y(y - 1) - 1(y - 1) = 0$

$(8y - 1)(y - 1) = 0$

$y = 1/8$ or $y = 1$

Put back $y = 2^x$

$2^x = 2^{-3}$ or $2^x = 2^0$

$x = -3$ or $x = 0$

Answer: Solution set = {−3, 0}

(v) $4^{1+x} + 4^{1-x} - 10 = 0$

Write $4^{1+x} = 4 \cdot 4^x$

Write $4^{1-x} = 4/4^x$

$4 \cdot 4^x + \frac{4}{4^x} - 10 = 0$

Let $4^x = y$

$4y + \frac{4}{y} - 10 = 0$

Multiply every term by y

$4y^2 - 10y + 4 = 0$

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Divide by 2

$2y^2 - 5y + 2 = 0$

$(2y - 1)(y - 2) = 0$

$y = 1/2$ or $y = 2$

$4^x = 1/2$

$2^{2x} = 2^{-1}$

$2x = -1$

$x = -1/2$

$4^x = 2$

$2^{2x} = 2^1$

$2x = 1$

$x = 1/2$

Answer: Solution set = {−1/2, 1/2}

(vi) $3^x + 3^{3-x} - 12 = 0$

Write $3^{3-x} = 27/3^x$

$3^x + \frac{27}{3^x} - 12 = 0$

Let $3^x = y$

$y + \frac{27}{y} - 12 = 0$

Multiply by y

$y^2 - 12y + 27 = 0$

$(y - 3)(y - 9) = 0$

$y = 3$ or $y = 9$

$3^x = 3^1$ or $3^x = 3^2$

$x = 1$ or $x = 2$

Answer: Solution set = {1, 2}

(vii) $5^{1+3x} + 5^{2-3x} - 126 = 0$

Write $5^{1+3x} = 5 \cdot 5^{3x}$

Write $5^{2-3x} = 25/5^{3x}$

$5 \cdot 5^{3x} + \frac{25}{5^{3x}} - 126 = 0$

Let $5^{3x} = y$

$5y + \frac{25}{y} - 126 = 0$

Multiply by y

$5y^2 - 126y + 25 = 0$

$5y^2 - 125y - y + 25 = 0$

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$5y(y - 25) - 1(y - 25) = 0$

$(5y - 1)(y - 25) = 0$

$y = 1/5 \quad \text{or} \quad y = 25$

$5^{3x} = 5^{-1} \quad \text{or} \quad 5^{3x} = 5^{2}$

$3x = -1 \quad \text{or} \quad 3x = 2$

$x = -1/3 \quad \text{or} \quad x = 2/3$

Answer: Solution set = {−1/3, 2/3}

(viii) $(x^2 + 1/x^2) + 2(x + 1/x) - 33 = 0$

Put $y = x + 1/x$

Squaring gives $x^2 + 1/x^2 = y^2 - 2$

$(y^2 - 2) + 2y - 33 = 0$

$y^2 + 2y - 35 = 0$

$(y + 7)(y - 5) = 0$

$y = 5 \quad \text{or} \quad y = -7$

Case $y = 5$

$x + 1/x = 5$

$x^2 - 5x + 1 = 0$

$x = (5 \pm \sqrt{21})/2$

Case $y = -7$

$x + 1/x = -7$

$x^2 + 7x + 1 = 0$

$x = (-7 \pm \sqrt{45})/2$

$= (-7 \pm 3\sqrt{5})/2$

Answer: $x = (5 \pm \sqrt{21})/2$, $x = (-7 \pm 3\sqrt{5})/2$

(ix) $2(x^2 + 1/x^2) - 25(x - 1/x) - 17 = 0$

Put $y = x - 1/x$

Squaring gives $x^2 + 1/x^2 = y^2 + 2$

$2(y^2 + 2) - 25y - 17 = 0$

$2y^2 - 25y - 13 = 0$

$2y^2 - 26y + y - 13 = 0$

$2y(y - 13) + 1(y - 13) = 0$

$(y - 13)(2y + 1) = 0$

$y = 13 \quad \text{or} \quad y = -1/2$

Case $y = 13$

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$x - 1/x = 13$

$x^2 - 13x - 1 = 0$

$x = (13 \pm \sqrt{173})/2$

Case $y = -1/2$

$x - 1/x = -1/2$

$2x^2 + x - 2 = 0$

$x = (-1 \pm \sqrt{17})/4$

Answer: $x = (13 \pm \sqrt{173})/2$, $x = (-1 \pm \sqrt{17})/4$

(x) $2x^4 - 5x^3 - 14x^2 - 5x + 2 = 0$

Divide every term by x^2

$2x^2 - 5x - 14 - 5/x + 2/x^2 = 0$

$2(x^2 + 1/x^2) - 5(x + 1/x) - 14 = 0$

Let $x + 1/x = y$

Then $x^2 + 1/x^2 = y^2 - 2$

$2(y^2 - 2) - 5y - 14 = 0$

$2y^2 - 5y - 18 = 0$

$2y^2 - 9y + 4y - 18 = 0$

$y(2y - 9) + 2(2y - 9) = 0$

$(2y - 9)(y + 2) = 0$

$y = 9/2$ or $y = -2$

Case $y = 9/2$

$x + 1/x = 9/2$

$2x^2 - 9x + 2 = 0$

$x = (9 \pm \sqrt{65})/4$

Case $y = -2$

$x + 1/x = -2$

$x^2 + 2x + 1 = 0$

$(x + 1)^2 = 0$

$x = -1$

Answer: $x = (9 \pm \sqrt{65})/4$, $x = -1$

(xi) $(x + 1)(x + 2)(x - 4)(x - 5) + 8 = 0$

Multiply the factors whose constant terms add to the same number

$(x + 1)(x - 4) = x^2 - 3x - 4$

$(x + 2)(x - 5) = x^2 - 3x - 10$

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Let $x^2 - 3x = y$

$(y - 4)(y - 10) + 8 = 0$

$y^2 - 14y + 40 + 8 = 0$

$y^2 - 14y + 48 = 0$

$(y - 6)(y - 8) = 0$

$y = 6$ or $y = 8$

Case $y = 6$

$x^2 - 3x - 6 = 0$

$x = (3 \pm \sqrt{33})/2$

Case $y = 8$

$x^2 - 3x - 8 = 0$

$x = (3 \pm \sqrt{41})/2$

Answer: $x = (3 \pm \sqrt{33})/2$, $x = (3 \pm \sqrt{41})/2$

(xii) $(x + 3)(x + 4)(x + 5)(x + 6) = -1$

$(x + 3)(x + 6) = x^2 + 9x + 18$

$(x + 4)(x + 5) = x^2 + 9x + 20$

Let $x^2 + 9x = y$

$(y + 18)(y + 20) = -1$

$y^2 + 38y + 360 = 0$

$(y + 18)(y + 20) = 0$

$y = -18$ or $y = -20$

Case $y = -18$

$x^2 + 9x + 18 = 0$

$(x + 3)(x + 6) = 0$

$x = -3, -6$

Case $y = -20$

$x^2 + 9x + 20 = 0$

$(x + 4)(x + 5) = 0$

$x = -4, -5$

Answer: Solution set = {−3, −4, −5, −6}

Remember: in each part, first put y equal to the repeated part, solve the quadratic in y, then go back to x. Reject any value that makes a denominator zero.

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